Brecht F. Verbeken

Quantitative economics | Data science | Artificial intelligence

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Profile

Quantitative economist (PhD) and mathematician (MSc) by training, working as a data scientist and researcher with strong interests in economics and AI. More recently, particularly interested in the economic impact of AI and its role in scientific discovery and innovation. Research experience in consumer-confidence nowcasting, model evaluation and AI in mathematics and science. Combines quantitative methods with institutional projects and reproducible analysis.

Appointments

Guest Professor (10%)

Vrije Universiteit Brussel | September 2025 - present

Course holder for Mathematics II for Business Economics (6 ECTS), with responsibility for teaching and assessment; master's-thesis supervision.

Postdoctoral Researcher (90%)

Vrije Universiteit Brussel | September 2024 - present

Research in model evaluation, AI capabilities and limitations, AI-assisted mathematics and stochastic systems. Institutional AI workflows, research proposal development and interdisciplinary collaboration.

Research Associate

Vrije Universiteit Brussel | September 2022 - September 2024

Uplift-model evaluation and semi-Markov processes; mathematics teaching, thesis supervision and assessment.

Doctoral Researcher with Teaching Duties

Vrije Universiteit Brussel | April 2018 - September 2022

Stochastic models for manpower planning; collaboration on economic nowcasting and ergodicity economics; teaching mathematics and manpower planning.

Education

PhD in Applied Economic Sciences - Business Engineering

Vrije Universiteit Brussel | 2024

On Models and Metrics: Improving Stochastic Model Performance in Manpower Planning.

Master of Mathematics - Fundamental Mathematics
Vrije Universiteit Brussel / University of Antwerp | 2018

Bachelor of Mathematics
Vrije Universiteit Brussel | 2016

Methods and tools

Mixed-frequency nowcasting, econometrics, statistical model evaluation, treatment-effect rankings, uncertainty, simulation, stochastic and semi-Markov processes. Python, R, LaTeX; textual analysis, APIs and local model workflows; reproducible analytical pipelines.

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Selected research and institutional work

Selected projects

AI and knowledge production. Interested in how AI changes the pace and direction of scientific discovery and innovation, the division of work between researchers and AI, and verification costs.

AI-supported science for policy. Through a VUB collaboration with the JRC, I was involved in developing evaluation templates and contributed to a validation protocol for AI-supported research outputs.

FARI / AI4Gov-X. Co-authored an internal review of AI4Scale requirements and prototype validation, covering feasibility, prioritisation, deployment, governance and process ownership.

Institutional AI adoption. Led technical implementation of LLM-assisted classification for an initial 4,684 course information sheets at VUB, with human validation and documented quality controls.

Public investment under uncertainty. Led scenario-based research on Brussels Metro 3, using official transport and open-map data to examine accessibility and geographical differences.

Mathematics. Research on stochastic matrices and spectral regions, including an illustrated Karpelevič project with Vincent Ginis. First author of a documented case study of AI-assisted mathematical proof development.

Selected publications and working papers

Probing the Trajectories of Reasoning Traces in Large Language Models
Marthe Ballon, Brecht F. Verbeken, Vincent Ginis, Andres Algaba (2026). *NeurIPS 2026 (accepted)*. Paper

Lexical hints of accuracy in LLM reasoning chains
Arne Vanhoyweghen, Brecht F. Verbeken, Andres Algaba, Vincent Ginis (2026). *Scientific Reports, 16, article 25593*. Paper

Early Evidence of Vibe-Proving with Consumer LLMs: A Case Study on Spectral Region Characterization with ChatGPT-5.2 (Thinking)
Brecht F. Verbeken, Brando Vagenende, Marie-Anne Guerry, Andres Algaba, Vincent Ginis (2026). *Working paper, arXiv:2602.18918*. Paper

Uplift Model Evaluation with Ordinal Dominance Graphs
Brecht F. Verbeken, Marie-Anne Guerry, Wouter Verbeke, Sam Verboven (2025). *Journal of Machine Learning Research, 26(90), 1-56*. Paper

Daily news sentiment and monthly surveys: A mixed-frequency dynamic factor model for nowcasting consumer confidence
Andres Algaba, Samuel Borms, Kris Boudt, Brecht F. Verbeken (2023). *International Journal of Forecasting, 39(1), 266-278*. Paper

Full bibliography, free paper versions and mathematical papers: All research

Teaching, service and languages

University-level teaching since 2018. Course responsibility for Mathematics II for Business Economics; supervision of more than 25 master's theses on economics and business topics; scientific writing and interdisciplinary collaboration.

Serve on PhD advisory committees, contributing to the assessment of doctoral research progress. HR and recruitment responsibilities at the Data Analytics Lab, including candidate selection.

Dutch: native. **English:** C2 listening/reading, C1 speaking/writing. **French:** C1 listening/reading, B2 speaking, B1 writing. **German:** A1-A2 by skill. CEFR levels are self-assessed.